

→ Ratio and proportion

$$A : B = 3 : 2 \rightarrow A = 3x, B = 2x$$
$$\begin{array}{ccc} \uparrow & & \uparrow \\ 9 & : & 6 \\ \uparrow & & \uparrow \\ 9 & : & 8 \end{array} = \frac{3:2}{3:2}$$

Banc

Ratio Proportion

$$A : B = 3 : 2$$
$$B : C = 5 : 4$$

$$A : B : C = 15 : 10 : 8$$

If $a : b = 7 : 9$ and $b : c = 15 : 7$, then what is $a : c$?

(a) 3 : 5

(b) 5 : 3

(c) 7 : 15

(d) 7 : 21

$$a : b = 7 : 9$$

$$b : c = 15 : 7$$

$$a : b : c = 105 : 145 : 63$$

$$a : c = 105 : 63$$

$$= 15 : 9$$

$$= \underline{\underline{5 : 3}}$$

If $A : B = 1/2 : 3/8$, $B : C = 1/3 : 5/9$ and $C : D = 5/6 : 3/4$ then the ratio $A : B : C : D$ is

(a) $4 : 6 : 8 : 10$

(b) $8 : 6 : 10 : 9$

(c) $6 : 8 : 9 : 10$

(d) $6 : 4 : 8 : 10$

$$A : B = \frac{1 \times 8}{2} : \frac{3 \times 8}{8} = 4 : 3$$

$$B : C = \frac{1}{3} : \frac{5}{9} = 3 : 5$$

$$C : D = \frac{5}{6} : \frac{3}{4} = 10 : 9$$

$$A : B : C : D = 120 : 90 : 150 : 135$$

$$= 40 : 30 : 50 : 45$$

$$= 8 : 6 : 10 : 9$$

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If $2A = 3B = 4C$, then $A : B : C$ is equal to

(a) $2 : 3 : 4$

(b) $3 : 4 : 6$

(c) $4 : 3 : 2$

(d) $6 : 4 : 3$

$$2A = 3B = 4C$$

$$A : B : C = \frac{12}{2} : \frac{12}{3} : \frac{12}{4}$$

$$= 6 : 4 : 3$$

The prices of a scooter and a television set are in the ratio of 3:2. If a scooter costs Rs 600 more than the television set, then the price of television set is:

- (a) Rs 1800
- (b) Rs 1200
- (c) Rs 2400
- (d) None of these

$$S : T = 3 : 2$$

$$\begin{array}{c} \uparrow \quad \quad \quad \downarrow \\ 1800 \quad \quad 1200 \\ \uparrow \quad \quad \quad \uparrow \\ S \quad \quad \quad T \end{array}$$

$$S = 3x$$

$$T = 2x$$

$$S = T + 600$$

$$3x = 2x + 600$$

$$x = 600$$

If an amount of Rs 1,50,000 is shared among A, B and C in the ratio of 2 : 3 : 5, then A receives the same amount as he would receive if another sum of money is shared between A, B and C in the ratio of 5 : 3 : 2. The ratio of Rs 1,50,000 to the second amount of money is

- (a) 2 : 3
- (b) 3 : 2
- (c) 5 : 2
- (d) 5 : 3

$$A : B : C = 2 : 3 : 5 = 30,000 : 45,000 : 75,000$$

$$\begin{array}{c} 15,00,000 \\ \uparrow \\ 1 = 15,000 \end{array}$$

$$1 = 60,000$$

$$A : B : C = 5 : 3 : 2 = 30,000$$

$$10 = 60,000$$

$$\frac{1,50,000}{60,000} = \frac{5}{2}$$



A mixture contains alcohol and water in the ratio of 12:5. On adding 14 litres of water, the ratio of alcohol to water becomes 4:3. The quantity of alcohol in the mixture is:

- (a) 18 litres
(b) 24 litres
(c) 26 litres
(d) None of these

$$\frac{A}{W} = \frac{12x}{5x + 14} = \frac{4}{3}$$

$$A = 12(x) = \frac{6}{1} \times \frac{7}{2} = 21$$

$$\Rightarrow 9x = 5x + 14$$

$$2x = 14$$

$$x = \frac{7}{2}$$

Salaries of A and B are in the ratio of 9 : 4 respectively. When A's salary is increased by 15%, it becomes Rs 5175. What is B's salary?

- (a) Rs 2000
(b) Rs 2500
(c) Rs 4000
(d) Rs 4500
(e) None of these

$$A = 9x \times 1.15 = 5175$$

$$\Rightarrow 10.35x = 5175$$

$$x = \frac{5175}{10.35} = 500$$

$$\frac{A}{B} = \frac{9x}{4x} = 2000$$

During our campaign against child labour we have found that in three glass making factories A, B and C there were total 63 children aged below 18 were involved. The ratio of male to female in A, B and C was 4 : 3, 3 : 2 and 5 : 4 respectively. If the no. of female children working in the factories B and C be equal then find the no. of female children working in factory A:

- (a) 5
(b) 2
(c) 8
(d) 6

	m	f
A	4x	3x
B	6y	4y
C	5z	4z

$3(2) = 6$
 $7x + 192 = 33$
 $\frac{7x}{2} + \frac{192}{2} = \frac{33}{2}$

$$2y = 42$$

$$y = 21$$

$$(x, 2) = \text{int}$$



Sometimes ago in a Cinema hall a blockbuster movie was being shown. Due to excessive demand of the show, manager of the cinema hall increased the prices of tickets in all the three categories. 25% in first class, 12.5% in special class and 40% in balcony. If the collection of one show was Rs 218925, then find the collection from the balcony only given that the ratio of price of ticket (increased price) for first class, special class and balcony is 5 : 9 : 14 :

- (a) 109462.5
(b) 58800
(c) 14400
(d) data insufficient

$$F : S : B = 5 : 9 : 14$$

$$\frac{\%}{\%} \rightarrow \text{perct}$$

$$2 \rightarrow 1/2$$

$$3 \rightarrow 2/3$$

$$5x + 9y + 14z = 218925$$

The Road Transport Corporation has hired a consultancy firm to determine the standard time that should be taken for a journey between two local destinations. The consultant after a detailed study and discussion submitted in his report that the time of journey is proportional to the product of the square root of number of passengers and the distance to be covered. For two distinct trips, the times taken are in the ratio of 2:1 and the distances covered are 80 km and 30 km, respectively. How many passengers are there in the first bus, if the second bus has 64 passengers?

- (a) 9
(b) 16
(c) 36
(d) 24

$$\begin{aligned} T_1 &\propto \sqrt{N_1} \cdot D_1 \\ T_2 &\propto \sqrt{N_2} \cdot D_2 \end{aligned} \quad \Rightarrow \frac{2}{1} = \frac{\sqrt{N_1} \cdot 80}{\sqrt{64} \cdot 30}$$

$$\begin{aligned} \frac{\sqrt{N_1} \cdot 80}{80 \cdot 3} &= \frac{1}{1} \\ \sqrt{N_1} &= 6 \\ N_1 &= 36 \end{aligned}$$

Between two stations Kanpur and Mumbai the first, second and third class fares were fixed in the ratio of 8:6:3. However, when Mr L Prasad, the railway minister, presented the railway budget, he reduced the first class fares by $\frac{1}{6}$ and the second class by $\frac{1}{12}$ in order to rationalize the fares vis-à-vis the second-class fares. In a year, ratio of the number of first, second and third class passengers were, respectively, 9:12:26 and the money at the booking offices was Rs1,08,800. How much was paid by the first class passengers?

- (a) Rs32,080
(b) Rs16,800
(c) Rs24,400
(d) None of these

$$F : S : T = \frac{8 \times 5}{163} : \frac{6 \times 11}{12 \times 2} : 3 = 40 : 33 : 18$$

$$F : S : T = 9 : 12 : 26$$

$$F : S : T = 360 : 396 : 468 = 1224$$

$$1 - \frac{1}{6} = \frac{5}{6} \quad \left| \quad 1 - \frac{1}{12} = \frac{11}{12} \right.$$

$$\begin{aligned} \frac{360}{1224} \times 1,08,800 \\ = \underline{\underline{32000}} \end{aligned}$$



Half of the volume of Petrol and kerosene mixture of ratio 7 : 5 is converted into a mixture of ratio 3 : 1 by the substitution (or replacement) method. While the mixture of ratio 7 : 5 was formed from the mixture of 7 : 3 by adding the Kerosene in it. If 240 litres petrol is required in the replacement method, what is the total amount of Kerosene was added to prepare the mixture of 7 : 5?

- (a) 100 litres
(b) 400 litres
(c) 50 litres
(d) 200 litres

$$P : K = 7 : 3 \rightarrow 5x - 3x = 2x = 100$$

$$P : K = 7 : (5 + k)$$

$$= 3 : 1$$

$$\frac{1}{2} = \frac{100}{x}$$

$$1 = 200$$

$$\frac{1}{x} = \frac{5}{12x} \left(1 - \frac{240}{12x}\right) \quad \left| \quad \frac{3}{5} = 1 - \frac{240}{12x} \right.$$

$$\frac{20}{x} = \frac{2}{5} \quad \boxed{x = 50}$$



In a mixture of petrol and Kerosene petrol is only 99 litres. If this same quantity of petrol would be presented in another mixture of petrol and Kerosene where total volume would be 198 litres less than the actual mixture then the concentration of petrol in the actual mixture would have been 13.33% point less than that of the new mixture. What is the concentration of petrol in actual mixture?

- (a) 20%
(b) 16.66%
(c) 26.66%
(d) 8.33%

	P	K	Total mix
(1)	99	x	99 + x

(2)	99	(x - 198)	x - 99
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$$\frac{99}{x - 99} \times 100 - \frac{99}{x + 99} \times 100 = 13.33$$

$$x = 396$$

$$\frac{x + 99 - x + 99}{x^2 - 99^2} \times 9900 = 13.33$$

$$\frac{(198) \times 9900}{40} = x^2 - 99^2$$



Mrs. Annapurna per day sells exactly four quintal sugar at Rs 2000 per quintal getting the profit of Rs 25%. Since she mixes two varieties of sugar, one costs Rs 14 per kg and another costs Rs 22 per kg. One day due to huge demand in market she had only 3 quintal of the required mixture so she purchased the sugar costs Rs 17 per kg at Rs 18 per kg from the wholeseller on that day and then she mixed 300 kg mixture with 100 kg sugar costing Rs 18 per kg to fulfill the demand of the market selling at the same price. How much percent less does she gain that she would have gained, if she had sufficient quantity of usual mixture of sugar?

- (a) 12.5%
(b) 18.18%
(c) 62.5%
(d) can't be determined

H.W
A.G.M

Hari and Murli have 24 cows and 30 cows respectively. Both of them together hired a grazing field for the whole month of November. Initially Hari's cows used for grazing then for the remaining days of the month Murli's cows grazed it. If Hari has paid Rs 3500 and Murli has paid Rs 5000 for grazing then for how many days Hari used the grazing field :

(a) 14

(b) 16

(c) 21

(d) 20

$$H = 24$$

$$M = 30$$

$$H = 3500 \Rightarrow H:M = 7:10$$

$$M = 5000$$

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$$\frac{24(x)}{30/x(30-x)} = \frac{7}{10}$$

$$24x = 630 - 21x$$

$$45x = 630 \Rightarrow \underline{\underline{x = 14}}$$